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Databricks & DevOps: **How Declarative Automation** **Bundles help with CI/CD**



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Cloud Data Architect, Data Strategist, Speaker, Blogger

Over 25 years of experience in IT

Founder of blog [AzurePlayer](#)

GitHub: **#adftools**, FabricTools and more...

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Former co-organiser of SQLDay (PL), Data Relay (UK),
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SQL, Data Platform, DevOps, Fabric, Databricks, AI
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@AzurePlayer.bsky.social



Agenda

- DevOps Fundamentals
- ~~Databricks Asset Bundles (DAB)~~
- Declarative Automation Bundles (DAB)
- Databricks GIT Folder

DevOps Fundamentals



DevOps Fundamentals



What is DevOps?

DevOps



Software Engineering
Best Practices



IT Operations



Fosters **collaboration**
between development and
operations teams to
automate Lakeflow Jobs
and **streamline processes**.

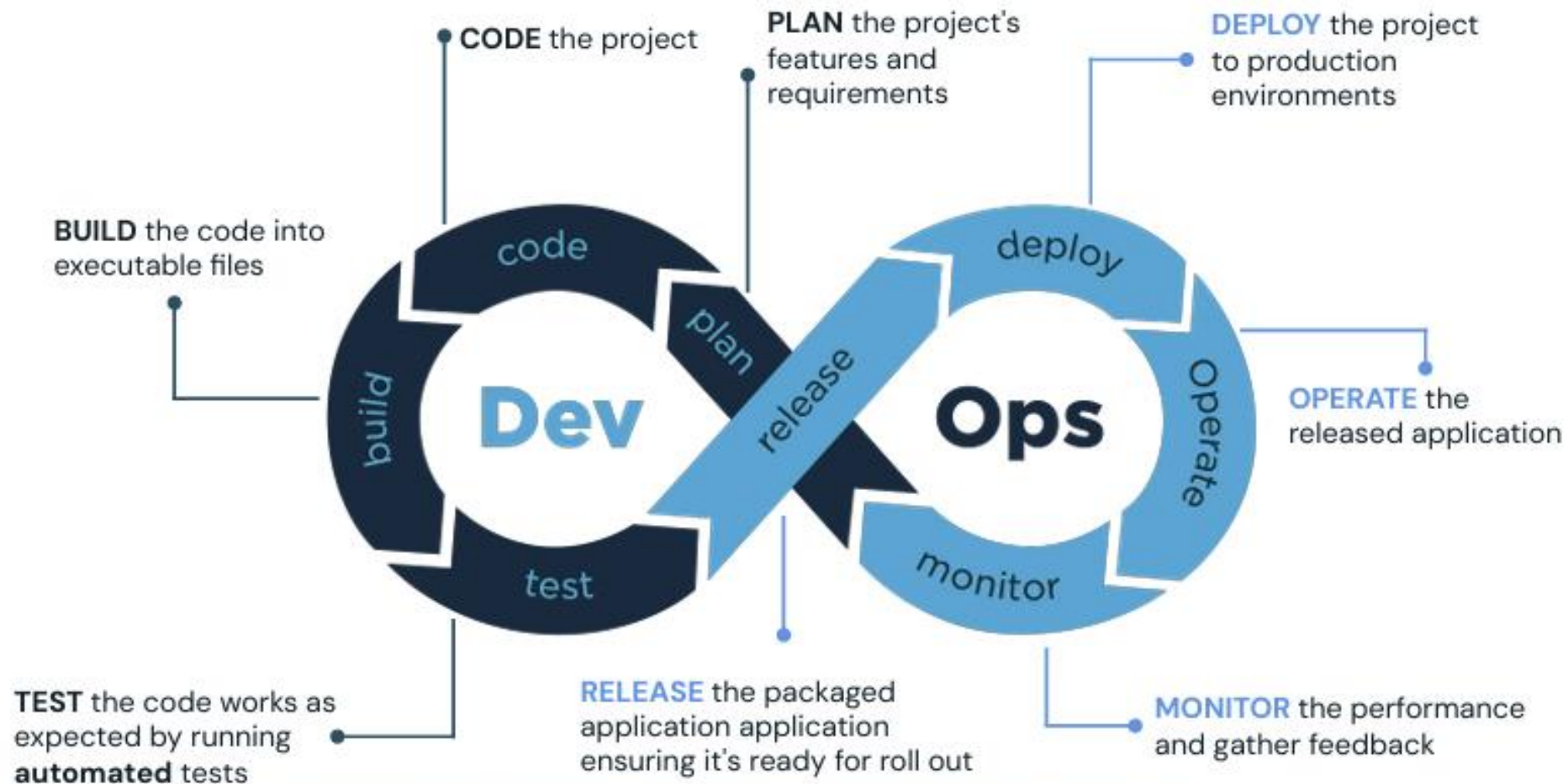


Key benefits include

- faster deployments
- improved collaboration
- enhanced reliability
- better scalability

DevOps Fundamentals

SDLC: 8 key steps



DevOps Fundamentals



DevOps, DataOps and MLOps

A set of practices, processes, and technologies

DevOps

**Software development
and IT operations**

- Automate CI/CD
- Enable continuous code testing
- Version control
- Establish Production-grade Lakeflow Jobs
- Orchestration & Automation
- Monitor system performance

DataOps

**Building quality data
pipeline processes**

- Optimize data processing
- Centralize data discovery, management, and governance
- Establish traceable data lineage and monitoring
- Enhance collaboration across teams
- Monitor data quality

MLOps

**ML model development
and deployment**

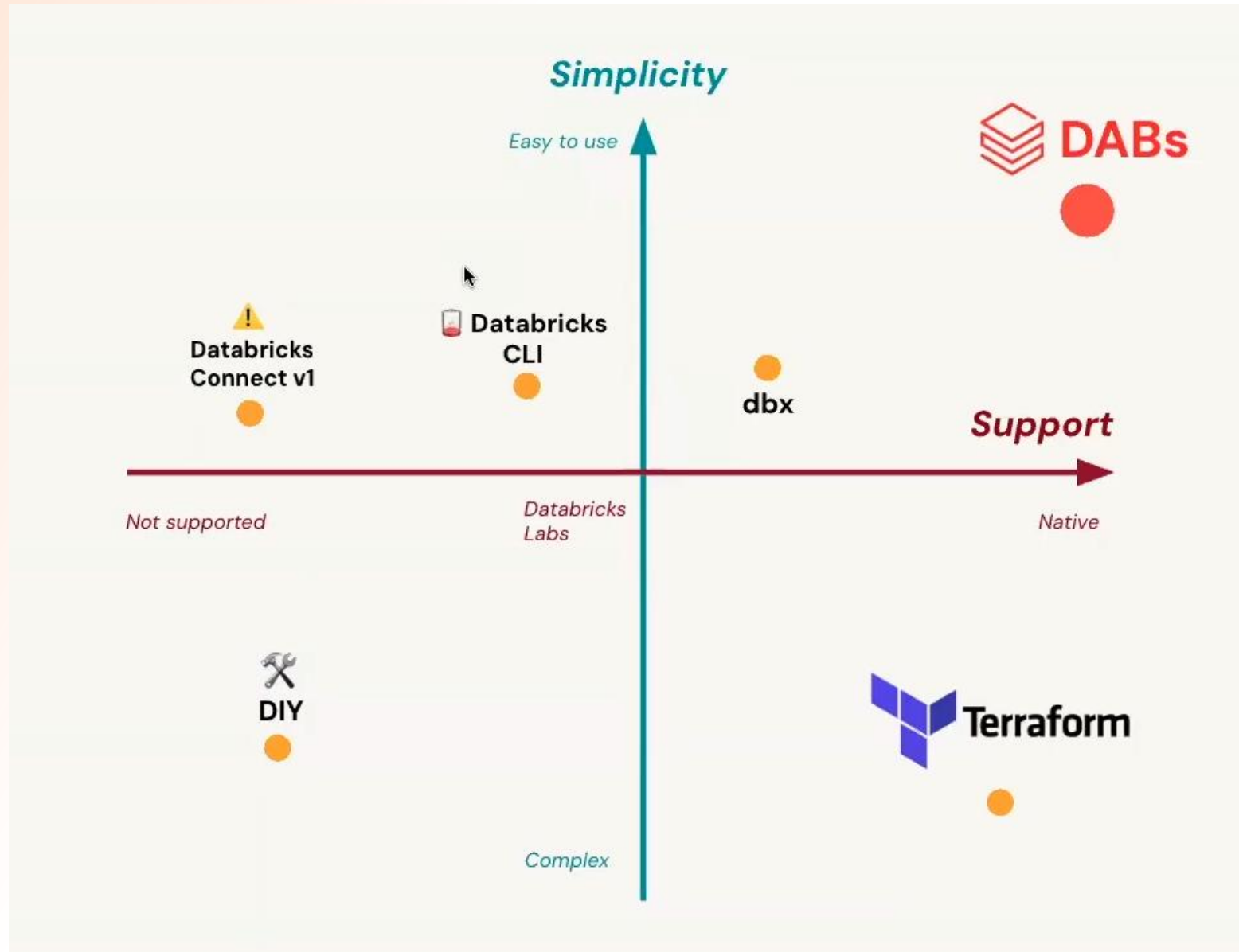
- Treating model code as software
- Treating models as data
- Manage the model lifecycle
- Monitor Model Performance



DAB

Declarative Automation Bundles
(Databricks Asset Bundles)

DABs is simple and native



Databricks CLI: Powerful Command Line Tool



- **Configure once**, reuse all devtools
- **Easy to use** – built in error handling, network retries, debug logging and secret redaction
- **Fully documented** with autocomplete, pagination and easy to read output

Databricks CLI: Install or Update



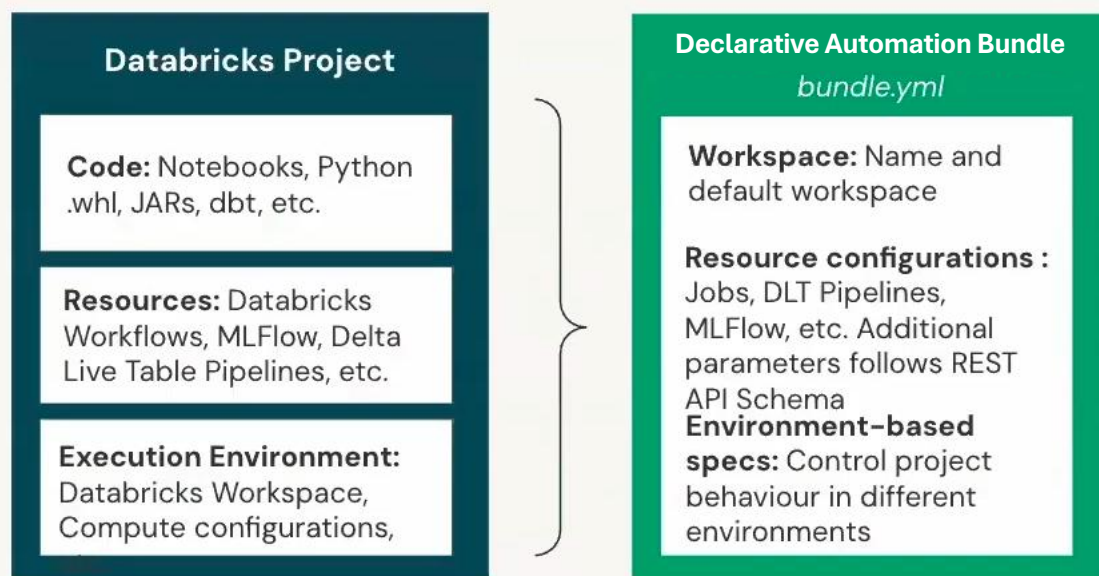
```
> winget search databricks  
> winget install Databricks.DatabricksCLI
```

What are Declarative Automation Bundles



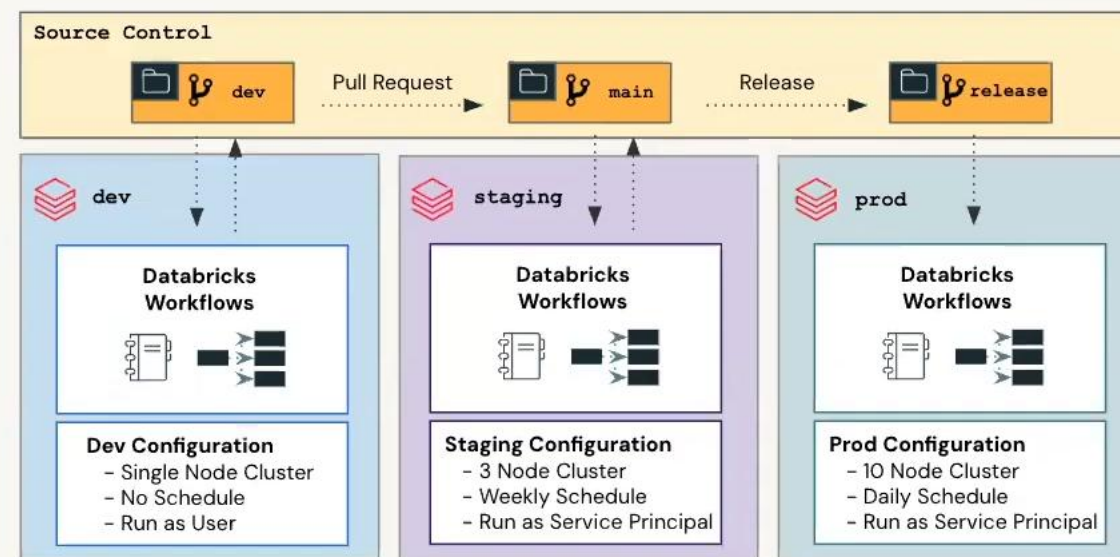
“Write once, deploy everywhere!”

Declarative Automation Bundles (DABs) are YAML files that specify the artifacts, resources and configurations of a Databricks project.



You can deploy to multiple workspaces, run your projects, tweak configs, and test changes from your IDEs, Terminal or from within Databricks!

You can **install the Databricks CLI** and begin to **deploy your bundles on your CI/CD server**; you can also **trigger automatically with Git Actions**



```
$ databricks bundle deploy -t "dev"
$ databricks bundle run pipeline --refresh-all -t "dev"
```

Bundle definition



Name and default
Workspace

Resource configurations

- Jobs, DLT pipelines, MLflow, etc.
- Follows REST API schema

Environment-based
specs

- Control project behavior in different environments

```
bundle.yml
1 bundle:
2   name: shark_sightings
3
4 workspace:
5   host: https://e2-dogfood.staging.cloud.databricks.com
6
7 resources:
8   jobs:
9     shark_sightings:
10      name: "${bundle.environment}] Shark sightings"
11      tasks:
12        - task_key: shark_sightings
13          notebook_task:
14            base_parameters:
15              dbname: "shark_sightings_${bundle.environment}"
16              notebook_path: ./shark_sightings.py
17            new_cluster:
18              spark_version: 10.4.x-scala2.12
19              num_workers: 1
20              node_type_id: i3.xlarge
21
22 environments:
23   development:
24     default: true
25   production:
26     workspace:
27       host: https://e2-demo-west.cloud.databricks.com
28     resources:
29       jobs:
30         shark_sightings:
31           schedule:
32             quartz_cron_expression: 14 8 14 * * ?
33             timezone_id: UTC
34
```

DABs demo



DEMO

DAB Project Structure

DABs' Options



Declarative Automation Bundles



Deployment Options

- **Local Deployment (IDE + CLI):**
Manual approach to locally deploy DABs for testing purposes.
- **Local Deployment with Git integration (IDE + Git Actions):**
Automated approach for deploying DABs with CI/CD best practices.
- **DABs at the workspace:**
Alternative to deploy and work on project development directly from the Databricks Workspace.
- **Workspace deployment with Git integration:**
Browser UI + Git Folder + Git Actions



1. Local Deployment (CLI)

databricks bundle init



`databricks bundle init`

It creates a new bundle project from a template. It's the starting point before you ever deploy or run.

When you run `databricks bundle init`, the CLI:

- **Asks you to pick a template.** The built-in ones include `default-python` (a Python project with a sample job and pipeline), `default-sql`, `dbt-sql`, and `mlops-stacks` (a full ML project structure). The exact list depends on your CLI version.
- **Asks a few questions**, such as the project name and whether to include a sample notebook, a Python package, or a pipeline.
- **Generates a project folder** containing:
 - `databricks.yml`, the main bundle config with `dev` and `prod` targets already set up,
 - a `resources/` folder with YAML definitions for jobs and pipelines,
 - a `src/` folder with sample code or notebooks,
 - often `tests/`, a `README.md`, and CI-friendly structure.

databricks bundle deploy



`Databricks bundle deploy -t dev`

- It deploys your Databricks Asset Bundle to the **dev** target environment.
- **databricks bundle deploy** reads your bundle config (databricks.yml plus any included YAML files) in the current project folder. It builds any artifacts (for example, Python wheels) and uploads your code and files to the workspace. Then it creates or updates the resources the bundle defines, such as jobs, pipelines, and ML models.
- **-t dev** (short for `--target dev`) picks the dev target from the targets: section of databricks.yml. A target decides which workspace to deploy to, which settings or variables to override, and the deployment mode.

If the `dev` target has `mode: development`, which is common, Databricks also:

- prefixes resource names with `[dev your_name]` so they don't collide with other people's deployments,
- pauses job schedules and triggers,
- deploys into your personal workspace folder.

databricks bundle run



```
databricks bundle run -t dev <job_name>
```

- It starts a job (or pipeline) from your bundle in the **dev** target environment.
- **databricks bundle run** triggers a run of a resource the bundle has already deployed.
- **-t dev** tells it to use the dev target, meaning the dev workspace and the dev-named copies of your resources.
- **<job_name>** is the resource's *key* from your YAML, not its display name.



2. Local Deployment with GIT Integration (Automated)

Option 2



- GitHub: https://github.com/NowinskiK/dabs_cicd_demo
- GitHub Action:
 - Deployment to Dev (dev.yaml) – only from “dev” branch
 - Deployment to Production (prod.yaml) – only from “main” branch



Connected to Declarative Automation Bundles [Learn more](#)

Runs **Tasks**



notebook_task

...dabs_project/prod/files/src/notebook
job_cluster

refresh_pipeline_2

Pipeline: demo_dabs_project_pipeline

Task name* ⓘ

notebook_task

Type*

Notebook

Source* ⓘ

Workspace

Path* ⓘ

...cloudformations.org/.bundle/demo_dabs_project/prod/files/src/notebook [🔗](#)

⚠ This task was deployed as part of a [bundle](#), avoid editing this copy, instead edit the source and redeploy the bundle. ✕

Compute* ⓘ

job_cluster

28 GB · 8 Cores · DBR 13.3 LTS · Spark 3.4.1 · Scala 2.12 [✎](#)

Depends on

Select task dependencies...

Dependent libraries ⓘ

+ Add





3. Workspace Deployment with GIT Integration (Automated)

Option 3



- Create GIT folder In Databricks Workspace
- Change object & push to the repo
- GitHub Action triggered & deploy to target (dev) env



4. DABs at the workspace

GIT Actions

Advantages of DABs at the workspace

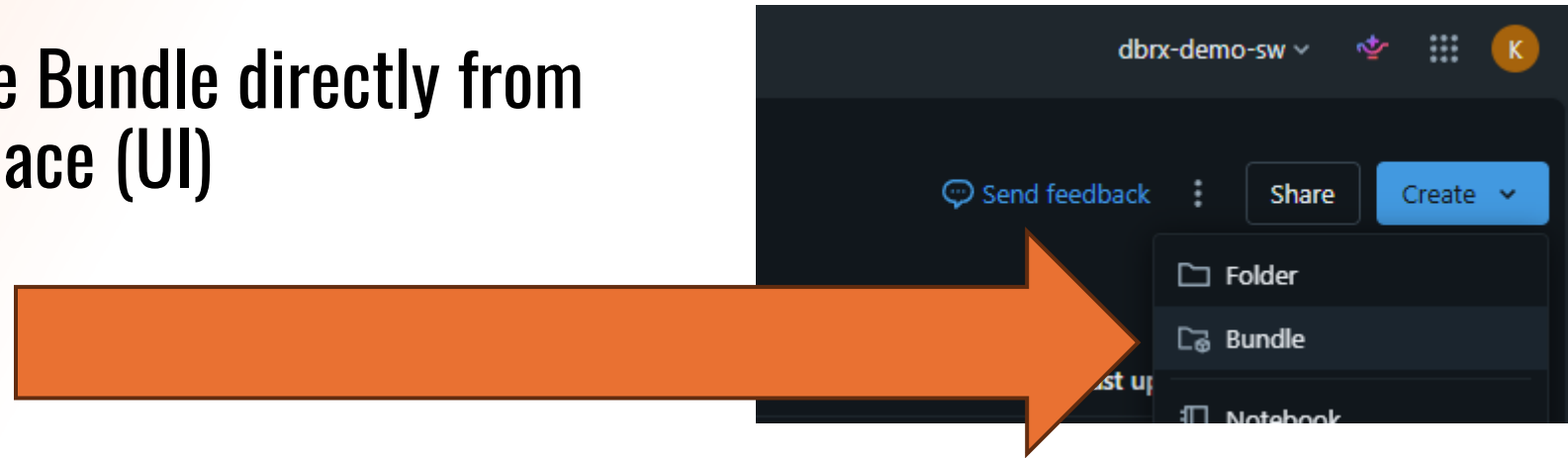


- Collaborate, develop and version directly on the Workspace, **no CLI needed.**
- **Collaborating on bundles directly in the workspace** allows for more rapid iteration and testing before moving to production.
- **You do not need to learn YAML** or know how to use the Databricks CLI to work with bundles in the workspace.

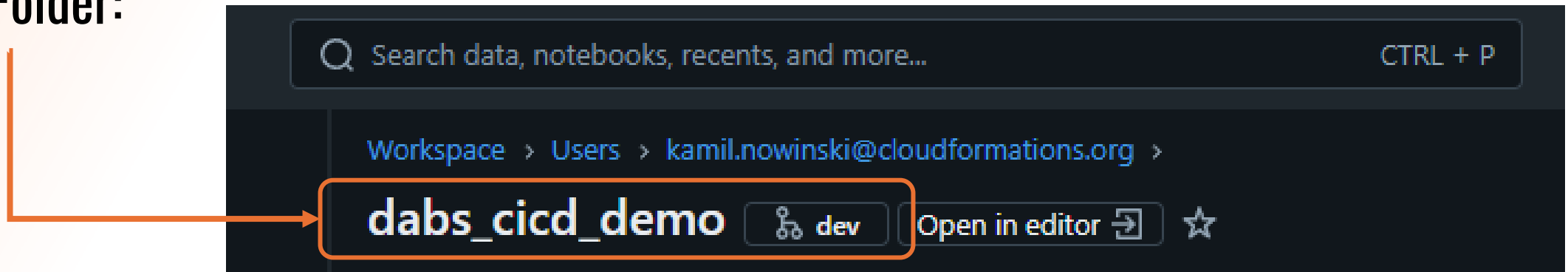
DABs at the workspace



- You can also create Bundle directly from Databricks Workspace (UI)



- The option is only available in context of GIT Folder:



DABs at the workspace



dabs_cicd_demo

dev

Open in editor

Search

Name

.git

.github

dabs_cicd_demo

demo_init

Open in bundle editor

dabs_cicd_demo

Deployments

Deployments (Ctrl + Alt + D)

Target: dev

Deploy Last deploy: unknown

Bundle resources

Jobs

[dev kamil_nowinski] demo_dabs_project_dbt_job

[dev kamil_nowinski] demo_dabs_project_job

[dev kamil_nowinski] Test job

Pipelines

[dev kamil_nowinski] demo_dabs_project_pipeline

Dashboards

[dev kamil_nowinski] NYC Taxi Trip Analysis

Apps

job-manager

New tab

Recents

Name

demo_dabs_project_job.yml

dabs_cicd_demo/resources

demo_dabs_project_pipeline.yml

dabs_cicd_demo/resources

README.md

demo_dabs_project

The 'demo_dabs_project' project was generated by using the default-

Getting started

1. Install the Databricks CLI from <https://docs.databricks.com/dev>

2. Authenticate to your Databricks workspace:



Bonus

Planning the Project

Setting Up Your Data Environments



Dev Data

- Often a Small Static Subset of Production Data
- Can be Anonymized or Synthetic Datasets
- Supports Rapid Development and Testing
- Ensures Privacy and Data Integrity



Stage Data

- Staging Data Mirrors Production Structure & Volume, Typically Static
- Can be Anonymized or Scrubbed Sensitive Information
- Ensures Realistic Testing and Validation



Prod Data

- Production Data: Live & Fully Operational
- Contains Real User Data
- Continuously Updated
- Requires High Security, Privacy, & Compliance Standards



References



References

- What are Declarative Automation Bundles?
- Install or update the Databricks CLI
- Declarative Automation Bundles resources

SQLDay Lite 2026 Gdańsk

Session feedback



Thank You!



Data
Community